

SIMATS ENGINEERING

ASSESSMENT TOOL 1: Agent Design Assessment

COURSE NAME: ARTIFICIAL INTELLIGENCE COURSE CODE: MLA0104
AND EXPERT SYSTEMS

COURSE OUTCOME COVERED

CO1: *Apply search algorithms and heuristic techniques to solve state-space search and optimization problems. (BTL 3)*

Assessment Tool Weightage: 55 %

Total Marks: 50

Time: 60 Mins

No. of Questions: 1

Annexure A Agent Design Assessment

Code of Conduct:

*I **P Dinesh Reddy / 192525399** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:

Question	Problem Understanding & AI Concept Identification	Dialogflow Agent Design & Implementation	Problem Formulation & Conversation Flow Design	Application of AI Techniques & Reasoning	Testing, Evaluation & Documentation	Total
Weightage	20%	30%	20%	20%	10%	100%
Q1						

Signature of the Faculty

Total Marks Awarded

Scenario

A university wants to develop an **AI-based Student Support Assistant** to provide instant responses to student queries related to course registration, timetable, examination schedules, attendance details, and campus services.

As an AI developer, design and implement a conversational agent using **Dialogflow** that can understand user requests, process intents, and provide appropriate responses.

AI-Based Student Support Assistant Using Dialogflow

Problem Statement

A university requires an AI-powered chatbot that provides students with instant responses to common academic and campus-related queries such as:

- Course registration
- Timetable
- Examination schedules
- Attendance details
- Campus services

The chatbot should understand natural language, identify the user's intent, and return accurate responses using Dialogflow.

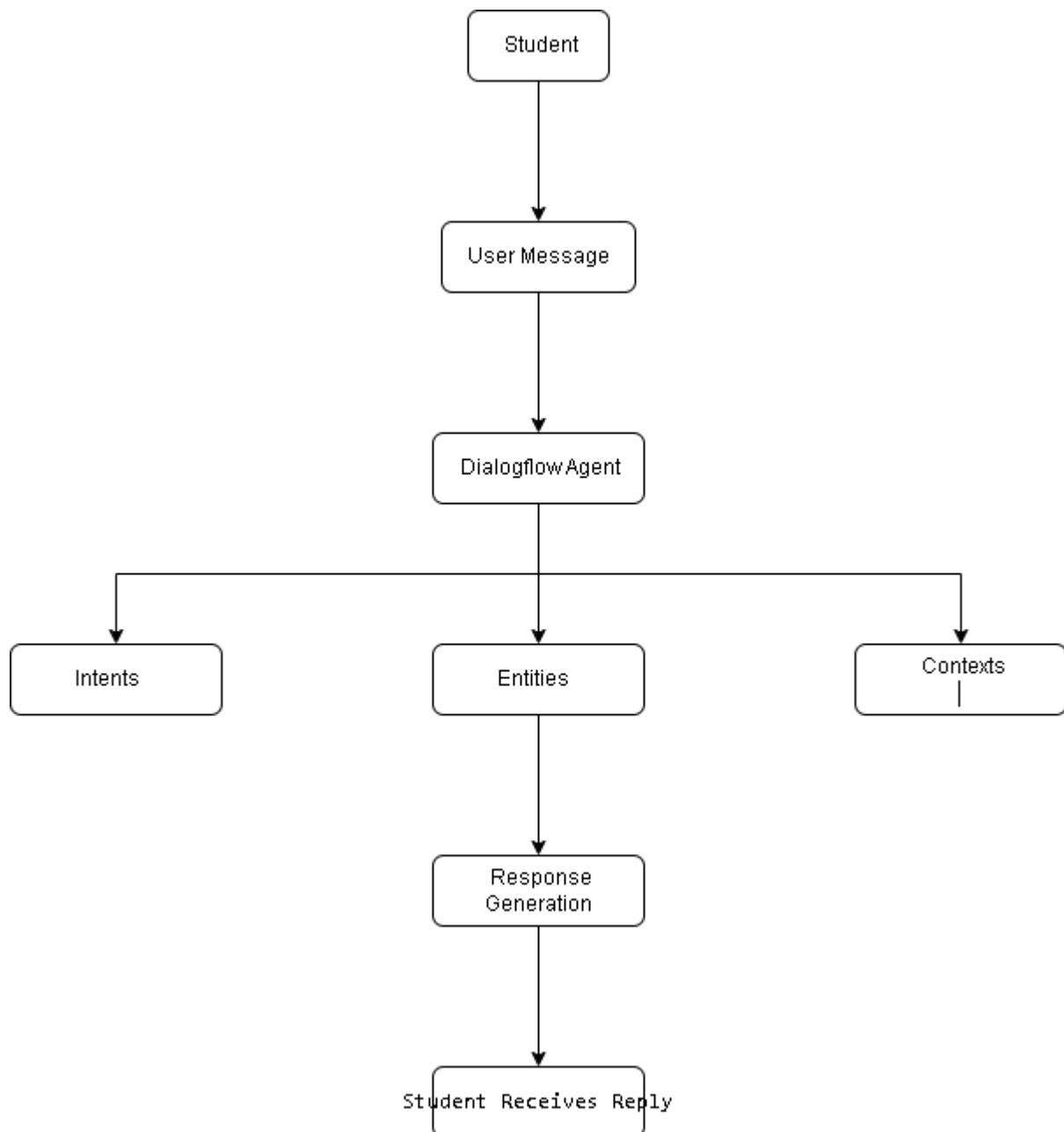
Objective

Develop a conversational AI using **Dialogflow** that can:

- Understand student queries
- Detect user intent
- Extract important information (entities)
- Respond with accurate information

- Reduce workload on university support staff
- Provide 24×7 assistance

System Architecture



Dialogflow Components

Agent Name

University Student Support Assistant

Default Language:

- English

Timezone:

- Asia/Kolkata

Intents

Intent 1: Course Registration

Training Phrases

- How do I register for courses?
- Register my course
- Course registration process
- Add a new subject
- Registration deadline

Response

Course registration can be completed through the University Student Portal.

Steps:

1. Login
2. Select Course Registration
3. Choose available subjects
4. Submit before the deadline.

For assistance contact the Academic Office.

Intent 2: Timetable

Training Phrases

- Show my timetable
- Today's classes
- Tomorrow timetable
- Class schedule
- When is my next class?

Response

Your class timetable is available in the Student Portal.

You can also download the weekly timetable from the Academic section.

Intent 3: Examination Schedule

Training Phrases

- Exam timetable
- Upcoming exams
- Exam dates
- Mid semester schedule
- Final exams

Response

The examination schedule has been published.

Visit the Examination Portal to download the latest timetable.

Good luck with your exams!

Intent 4: Attendance

Training Phrases

- My attendance
- Attendance percentage
- Attendance details
- How much attendance do I have?
- Check attendance

Response

Attendance information is available in the Student ERP Portal.

Students must maintain at least 75% attendance to appear for examinations.

Intent 5: Campus Services

Training Phrases

- Library timings
- Hostel office
- Bus facility
- Cafeteria
- Medical center
- WiFi support

Response

Available Campus Services:

- Library
- Hostel
- Transport

- Medical Center
- Cafeteria
- WiFi Help Desk
- Placement Cell

Please specify the service you need.

Intent 6: Greeting

Training Phrases

- Hi
- Hello
- Good morning
- Hey

Response

Hello!

Welcome to the University Student Support Assistant.

How may I help you today?

Intent 7: Goodbye

Training Phrases

- Bye
- Thank you
- Goodbye
- See you

Response

Thank you for using the Student Support Assistant.

Have a great day!

Entities

Entity	Example Values
@course	AI, ML, Python, DBMS
@department	CSE, AI&DS, ECE, EEE
@semester	Semester 1, Semester 2, Semester 3
@service	Library, Hostel, Transport, Medical

Contexts

Context	Purpose
Registration	Handles follow-up registration questions
Exam	Maintains examination conversation
Attendance	Handles attendance-related queries
Services	Provides additional campus service information

Sample Conversation

Student: Hi

Bot: Hello! Welcome to the University Student Support Assistant. How may I help you today?

Student: How do I register for courses?

Bot: Login to the Student Portal → Course Registration → Select subjects → Submit before the deadline.

Student: Show my timetable

Bot: Your timetable is available in the Student Portal under Academic Schedule.

Student: What is my attendance?

Bot: Attendance information is available in the ERP Portal. Students must maintain at least 75% attendance.

Student: Where is the medical center?

Bot: The Medical Center is located near the Administrative Block and is open from 9:00 AM to 5:00 PM.

Student: Bye

Bot: Thank you for using the Student Support Assistant. Have a great day!

Workflow

Start



Student asks a question



Dialogflow receives input



Intent Matching



Entity Extraction



Generate Response



Display Reply



End

Advantages

- 24×7 student support
- Instant responses
- Reduces administrative workload
- Improves student experience
- Easy to update with new information
- Supports natural language conversations

Future Enhancements

- Integration with the university ERP system
- Real-time attendance retrieval
- Personalized timetable display
- Exam result notifications
- Voice-enabled chatbot
- Multilingual support (English, Tamil, Telugu, Hindi)

Conclusion

The **AI-Based Student Support Assistant** developed using **Dialogflow** provides an efficient and user-friendly solution for handling common student queries. By leveraging intents, entities, and conversational contexts, the chatbot delivers accurate responses regarding course registration, timetables, examination schedules, attendance, and campus services. This system

enhances the student experience by offering fast, reliable, and 24/7 support while reducing the workload on university administrative staff.

Q. No.	Criteria to be evaluated	BL	Marks
1	Problem Analysis and Agent Design: Analyze the student support problem and explain how Artificial Intelligence can solve the problem. Design the conversational agent by defining its objective, users, environment, inputs, outputs, and expected performance measures.	BL3	10
2	Dialogflow Agent Development: Create a Dialogflow agent by designing appropriate Intents, Entities, Training Phrases, Responses, and Contexts. Explain how the agent perceives user queries and generates responses. Provide screenshots of the implementation.	BL5	10
3	Problem Formulation and Decision Flow: Represent the chatbot interaction as a problem-solving process by defining the initial state, user goal, possible actions, state transitions, and goal completion conditions. Draw the conversation flow diagram.	BL6	10
4	Search Strategy Application: Analyze how search techniques used in AI problem solving can support conversational agents. Apply a suitable search approach (such as Breadth First Search or heuristic-based search) to model intent selection or dialogue navigation and justify your choice.	BL5	10
5	Testing and Evaluation: Test the Dialogflow agent with different user queries. Evaluate accuracy, response quality, handling of unknown queries, and suggest improvements.	BL6	10

Evaluation Rubrics

Criteria	Weightage (%)	Excellent (Full Marks)	Good	Average	Poor
Problem Understanding & AI Concept Identification	20%	10 – Clearly analyzes the student support problem, defines AI application, agent objective, users, environment, inputs, outputs, and performance measures with proper justification.	7.5 – Identifies most components of the agent design with minor omissions.	5 – Provides basic problem analysis with limited explanation of agent components.	0–2.5 – Incomplete understanding of problem and agent design.
Dialogflow Agent Design & Implementation	30%	10 – Successfully creates a functional Dialogflow agent with appropriate intents, entities, training phrases, responses, contexts, and explains interaction flow with screenshots.	7.5 – Develops major Dialogflow components with minor configuration errors.	5 – Creates basic intents and responses with limited entity/context utilization.	0–2.5 – Incomplete or incorrect Dialogflow implementation.
Problem Formulation & Conversation Flow Design	20%	10 – Clearly represents chatbot interaction using initial state, goal state, actions, state transitions, goal conditions, and a well-designed conversation flow diagram.	7.5 – Provides a structured flow with minor missing details.	5 – Provides basic flow representation with limited explanation.	0–2.5 – Poor or missing problem formulation and decision flow.
Application of AI Techniques & Reasoning	20%	10 – Correctly analyzes and applies suitable search strategies (BFS/heuristic search) for intent selection or dialogue navigation with strong justification.	7.5 – Applies appropriate search strategy with reasonable explanation.	5 – Provides basic understanding of search strategy application.	0–2.5 – Unable to relate search concepts with conversational agent.
Testing, Evaluation & Documentation	10%	10 – Performs comprehensive testing using multiple user queries, evaluates accuracy, response quality, unknown query handling, and suggests meaningful improvements.	7.5 – Performs adequate testing and provides evaluation with minor gaps.	5 – Conducts limited testing with basic observations.	0–2.5 – Insufficient testing and evaluation.